

Aitezaz Sikandar Khan

Full - Stack Developer

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Professional Summary

Full-Stack Software Engineer specializing in building modern web applications, real-time architectures, and cloud-deployed services using TypeScript, React, Next.js, and Node.js. Hands-on expertise across MongoDB database design, REST API engineering, WebSockets (Socket.io), third-party integrations (Stripe, LLMs), and containerized deployments with Docker and AWS EC2.

Education

University of Peshawar

Peshawar, Pakistan

Bachelor of Science in Computer Science (GPA: 3.60 / 4.00)

2022 -- 2026

- **Achievements:** Secured the 95th percentile nationwide in the HEC National Skills Competency Test (NSCT) organized by HEC, PSEB, and P@SHA (April 2026).
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Computer Networks, Software Engineering

Technical Skills

Languages: JavaScript, TypeScript, HTML5, CSS3, SQL

Frontend: React.js, Next.js, Redux Toolkit, Tailwind CSS, Material UI (MUI), Responsive Design

Backend: Node.js, Express.js, RESTful APIs, REST API Design, WebSockets (Socket.io)

Databases: MongoDB, Mongoose ODM, MySQL, Firebase

DevOps & CI/CD: Docker, GitHub Actions, AWS EC2, Nginx, Linux

Tools & Practices: Git, GitHub, Postman, Jest, Zod, Stripe API, Cloudinary, Agile/Scrum

Professional Experience

Full-Stack Software Engineer

Software Development

2025 -- Present

- Delivered 3 production MERN-stack applications end-to-end for local business clients (a complaint management portal, a personal finance ledger, and a content blog), handling requirements gathering, database design, deployment, and ongoing maintenance.
- Built a complaint tracking system that reduced manual follow-up by automating email notifications (Nodemailer), supporting media attachments (Cloudinary), and routing tickets via priority-based task queues.
- Implemented role-based access control with JWT authentication and Bcrypt hashing, protecting 15+ API endpoints and ensuring data isolation between user roles.

Projects

Collaborative Study Platform | React, Socket.io, Node.js, Gemini & Groq

- Developed a real-time virtual classroom utilizing Socket.io to synchronize slide changes, active users, and typing indicators with sub-100ms latency for up to 50 concurrent students.
- Designed a multi-LLM fallback mechanism that automatically redirects requests from Groq (Llama 3) to Google Gemini (3.5-Flash) when rate limits are hit, preventing AI chat interruptions.
- Eliminated audio latency by streaming text-to-speech voice narration directly to the browser using Express chunked transfer encoding headers.

Hospital Management System | React, Node.js, Express, MongoDB, Gemini API

- Built a secure, multi-role portal (Admin, Doctor, Patient) with appointment booking, using moment.js validation to prevent scheduling conflicts and Nodemailer for automated email reminders.
- Integrated an AI symptom-analysis assistant via the Gemini API, structuring prompt fallbacks to summarize patient history and generate relevant questions for doctor visits.
- Designed MongoDB aggregation pipelines to calculate monthly appointment distributions and doctor earnings, rendering the data on interactive dashboard charts.

DevOps-Enabled Blog Platform | React, Node.js, Docker, GitHub Actions, AWS EC2

- Packaged the React frontend and Node.js backend into multi-stage Docker containers, reducing image sizes via alpine bases to optimize deployment speed.
- Automated the deployment pipeline by writing a GitHub Actions workflow that builds Docker images and redeploys them to AWS EC2 via secure SSH commands.
- Configured Nginx as a reverse proxy to route public traffic, handle backend API requests, and serve compressed assets for reduced page load times.

E-Commerce Platform | Next.js, TypeScript, Redux, Stripe API, MongoDB

- Developed a fully typed storefront using Next.js App Router and TypeScript, managing persistent shopping cart states across pages with Redux Toolkit.
- Integrated Stripe Checkout for secure payments, implementing a backend webhook route to save orders to MongoDB and reset customer carts upon payment verification.
- Secured checkout forms and API endpoints against invalid data and malicious inputs using schema-based Zod validation and React Hook Form.